# Research Note 6

## 1 The difference of v scaling

First I rederived my results by hand for Process 1 and it correspond with B2,B3,B4.
Then, the paper 1302.3898 page 2 says for coulomb limit $R >> 1$, the scaling for v is always $\sigma \propto v^{-4}$. However my result is $v^{-2}$. (The paper mentioned a $v^{-2}$ scaling but is for strong interaction, quantum resonant regime, not our case).
Even if I use $M_A = \frac{m_\chi v}{R}$ to substitute $M_A$ to get $\sigma \propto v^{-4}$, there is still $R^2$ correspondence remained which is v dependence. Since $R >> 1$, this will cause $\sigma$ diverge, and $\alpha_\chi \approx \frac{m_\chi^{\frac{3}{2}}}{R} \to 0$.

Nevertheless, use $\sigma_V$ will give the predicted v scaling for $R >> 1$. In conclusion, the scaling $v^{-2}$ in my result is not the physical essence, it is a incorrect mathematical processing method.

## 2 Modification of Scattering Cross-Sections

To characterize dark matter self-interactions effectively, we modify the standard cross-section by introducing an angular weighting function $F(\theta)$. The standard transfer cross-section $\sigma_T$, defined with $F(\theta) = 1 - \cos\theta$, is designed to regulate forward scattering divergences ($\theta \to 0$). However, it fails for identical particles (as seen in Process 2: $\chi\chi \to \chi\chi$) because it cannot account for the backward divergence at $\theta = \pi$.

Notice that using the viscosity cross section $\sigma_V$ with $F(\theta) = \sin^2\theta$ resolves this issue by regulating both forward and backward divergences and it respect $\theta$ symmetry as $1 - cos\theta$ did.

## 3 something needs to take care of

Will using $\sin^2\theta$ omit the physical information contained in $1 - cos\theta$, and will using $1 - cos\theta$ omit the physical information contained in $\sin^2\theta$?

### 3.1 $1 - cos\theta$ is more fundamental

First I consider $1 - cos\theta$ is more fundamental. By noting the identity:

$$\sin^2\theta = 1 - \cos^2\theta = (1 - \cos\theta)(1 + \cos\theta) \tag{1}$$

we can interpret the viscosity weight as a fundamental extension of the transfer weight. The term $(1-\cos\theta)$ provides the base regulation for forward scattering. The additional term $(1 + \cos\theta)$ acts as a corrective factor:

- At $\theta = 0$, it is equal to 2, which does not interfere with the zeroing effect of the first term.
- At $\theta = \pi$, it is equal to 0, successfully eliminating the spurious factor of 2 that would otherwise arise from the $(1 - \cos\theta)$ term in a region where identical particle indistinguishability should be enforced.

It can be considered that $(1 + \cos\theta)$ mathematically completely fills the hole left by $(1-\cos\theta)$, and also successfully compensates for the possible damage to the original physics that might result from the addition of this extra term. Or at least highlights the dominants and covers the minors as I will mention later. In this case, the other expressions in form $(1 - cos\theta) * f(\theta)$ should be possible, which is used to describe other scenarios. However, it is necessary to ensure that theta possesses the appropriate symmetry in those situations.

### 3.2 $sin^2\theta$ is more fundamental

$sin^2\theta$ is exactly $1 - cos^2\theta$, then we can consider $F(\theta)$ to follow a general form of $1 - \cos^n\theta$. It immediately remind me of Legendre polynomials $P_l(\cos\theta)$.

- Transfer Cross-Section ($\sigma_T$): $n = 1$
- Viscosity Cross-Section ($\sigma_V$): $n = 2$ .

if consider Legendre polynomial,

- Standard Cross-Section ($\sigma$): $F = P_0$ .
- Transfer Cross-Section ($\sigma_T$): $F = 1 - P_1 = P_0 - P_1$ .
- Viscosity Cross-Section ($\sigma_V$): $F = \frac{2}{3}(1 - P_2) = \frac{2}{3}(P_0 - P_2)$.

Higher-order terms $a(1 - P_n) = a(P_0 - P_n)$ could describe following scattering dynamics. While these Fs are orthogonal.

## 4 Why we should consider $\sigma_T$ & $\sigma_V$

- **Transfer Cross-Section ($\sigma_T$):** The transfer cross section is actually the integration of (the probabilities of particles scattering in all directions multiplied by the ratio of the corresponding momentum loss in the

z-direction). That is to say, we are considering how much longitudinal momentum (along the direction of $\vec{p}_{in}$ has been lost. And the loss of longitudinal momentum will cause the radius of the halo to decrease.

- **Viscosity Cross-Section ($\sigma_V$):** The viscosity cross section begins to emphasize the gain of transverse momentum (in the direction perpendicular to $\vec{p}_{in}$. I think this will cause the shape of the halo to become more divergent.

However, what is the relationship with these above and our goal to determine coupling constants?

This might be a problem on the experimental observation. The experiment gives the SIDM limit is about $\frac{\sigma_T}{m_\chi} \leq 1 \frac{cm^2}{g}$. But didn't give the limit for $\sigma_V$. So we have to have a new limit by experiment for $\sigma_V$.

# 5 general method

Following the approach in the paper 1302.3898, they were using a method called: partial wave analysis. Using their own words, "to examine the angular dependence of the scattering cross section in the classical and strongly-coupled regimes, observing the transition to the weakly coupled regime with forward-peaked Rutherford scattering. " I guess in this method people can get the correct $\frac{d\sigma}{d\Omega}$. However we still need to know $F(\theta)$. For different angular distribution and its influence on the structure of halo, there are different modification to use. $1 - cos\theta$ & $sin^2\theta$ are two of them. But for some angle distribution, $1 - cos^8\theta$ may be a better one. To determine more accurate result, there is a method called N-body simulation, which inputs physics laws and collision rules, and output the formation of haloes. If people use an N-body simulation to get a ground truth formation, and use different modification candidates to do variation. There will emerge one best modification. While $\sigma_T$ and $\sigma_V$ are people trying to grasp the important things and give up trivial information. Serving as excellent approximations for rapid "sanity checks" and extract dominant physical behaviors.

# 6 A conclusion

The bound $\sigma/m \lesssim 1\ \mathrm{cm}^2\,\mathrm{g}^{-1}$ represents the maximum allowed change in the structure of the dark matter halo. Here, $\sigma$ is the effective cross section, which can be understood as the product of the average change in the halo's structure per collision and the total number of collisions. Using the ordinary total scattering cross section yields one result, but for different scattering angular distributions, we need other modified cross sections, such as the transfer cross section $\sigma_T$ and the viscosity cross section $\sigma_V$. For these modified cross sections, the maximum allowed change in the halo structure remains the same, but the

average change per collision and the number of collisions that can occur are different. For different physical processes with different angular distributions, we adopt different modifications. The relationship between each modified cross section and the coupling strength also differs, leading to different constraints on the coupling strength.

The transfer cross-section is most appropriate when most particles are back-scattered, while the viscosity cross section is best when most particles are transversely scattered. For other angular distributions, even more tailored modified cross sections may exist. The transfer cross-section was introduced to quantify momentum loss in the incident direction, because greater momentum loss in this direction has a larger effect on the halo structure, while scattering with almost no longitudinal momentum loss leaves the halo essentially unchanged. For identical particle scattering, forward and backward scattering are indistinguishable, and these angles do not change the net incident momentum, thus not affecting the halo structure. The $\sin^2\theta$ factor in the viscosity cross section precisely satisfies this requirement: it vanishes at $\theta = 0$ and $\theta = \pi$.

The natural question is: why $\sin^2\theta$, and not some other function $F(\theta)$ that also vanishes at those angles, such as $\sin^4\theta$? The answer may lie in the fact that we are interested in the transverse deformation of the dark matter halo, which is described by a tensor's traceless part. We start with the momentum flux density tensor per unit mass, which is the second moment of velocity, $v_i v_j$. We subtract the diagonal term $\frac{1}{3}\delta_{ij}\mathrm{Tr}(v_i v_j)$ to obtain the traceless velocity tensor

$$Q_{ij} = v_i v_j - \frac{1}{3}\delta_{ij}|\mathbf{v}|^2.$$

This tensor exists both before and after a collision. According to *the Mathematical Theory of Non-Uniform Gases* (by Sydney Chapman, T. G. Cowling, page 87, formula 1, F is Q, $(F + F1 - F' - F1')^2$ is the double dot product, which is the Frobenius norm square. Which is actually the Boltzmann H theorem. This means the contribution of a single collision to the macroscopic viscous dissipation is proportional to the change in this particle's tensor, specifically the square of the Frobenius norm of the change:

$$\Delta Q^2 = \sum_{i,j} \left(Q'_{ij} - Q_{ij}\right)^2.$$

We now perform the tensor calculation in the center-of-mass frame. Let the initial velocity of particle 1 be $\mathbf{v} = (0, 0, a)$, where $a = g/2$ and $g$ is the relative speed. Its initial traceless tensor is

$$Q^{(0)} = a^2 \begin{pmatrix} -\frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & \frac{2}{3} \end{pmatrix}.$$

After an elastic collision, the velocity becomes

$$\mathbf{v}' = a(\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta),$$

and the final traceless tensor $Q'$ has components

$$Q'_{ij} = v'_i v'_j - \frac{1}{3}\delta_{ij} a^2.$$

Explicitly,

$$Q' = a^2 \begin{pmatrix} \sin^2\theta\cos^2\phi - \frac{1}{3} & \sin^2\theta\cos\phi\sin\phi & \sin\theta\cos\theta\cos\phi \\ \sin^2\theta\cos\phi\sin\phi & \sin^2\theta\sin^2\phi - \frac{1}{3} & \sin\theta\cos\theta\sin\phi \\ \sin\theta\cos\theta\cos\phi & \sin\theta\cos\theta\sin\phi & \cos^2\theta - \frac{1}{3} \end{pmatrix}.$$

We compute $\Delta Q^2 = \|Q' - Q^{(0)}\|_F^2$ using the identity

$$\|Q' - Q^{(0)}\|_F^2 = \mathrm{Tr}\big((Q' - Q^{(0)})^2\big) = \mathrm{Tr}(Q'^2) + \mathrm{Tr}(Q^{(0)2}) - 2\,\mathrm{Tr}(Q' Q^{(0)}).$$

Because $Q'$ is just a rotated version of $Q^{(0)}$, both have the same Frobenius norm:

$$\mathrm{Tr}(Q'^2) = \mathrm{Tr}(Q^{(0)2}) = a^4\left(\frac{1}{9} + \frac{1}{9} + \frac{4}{9}\right) = \frac{2}{3}a^4.$$

The only angle-dependent term is $\mathrm{Tr}(Q'Q^{(0)})$. Since $Q^{(0)}$ is diagonal, only the diagonal elements of $Q'$ contribute:

$$\mathrm{Tr}(Q'Q^{(0)}) = a^4\left[\left(-\frac{1}{3}\right)\left(\sin^2\theta\cos^2\phi - \frac{1}{3}\right) + \left(-\frac{1}{3}\right)\left(\sin^2\theta\sin^2\phi - \frac{1}{3}\right) + \left(\frac{2}{3}\right)\left(\cos^2\theta - \frac{1}{3}\right)\right].$$

Using $\cos^2\phi + \sin^2\phi = 1$, this simplifies to

$$\mathrm{Tr}(Q'Q^{(0)}) = a^4\left[-\frac{1}{3}\sin^2\theta + \frac{2}{3}\cos^2\theta\right] = a^4\left(\cos^2\theta - \frac{1}{3}\right).$$

Substituting back,

$$\Delta Q^2 = \frac{2}{3}a^4 + \frac{2}{3}a^4 - 2a^4\left(\cos^2\theta - \frac{1}{3}\right) = 2a^4 - 2a^4\cos^2\theta = 2a^4\sin^2\theta.$$

Since $a = g/2$, we have $a^4 = g^4/16$, and therefore

$$\Delta Q^2 = \frac{g^4}{8}\sin^2\theta = 2\left(\frac{g}{4}\right)^2\sin^2\theta.$$

Hence we have $\sin^2\theta$ as the modification of angular distribution. While $(\frac{g}{4})^2 * 2$ is not related to angle, the viscosity cross section itself just needs the angle modification geometrically.

However, I don't think this is a satisfying explanation for why it is $sin^2\theta$ instead of $Const * sin^2\theta$. Maybe trying to fit the experimental result is one reason.